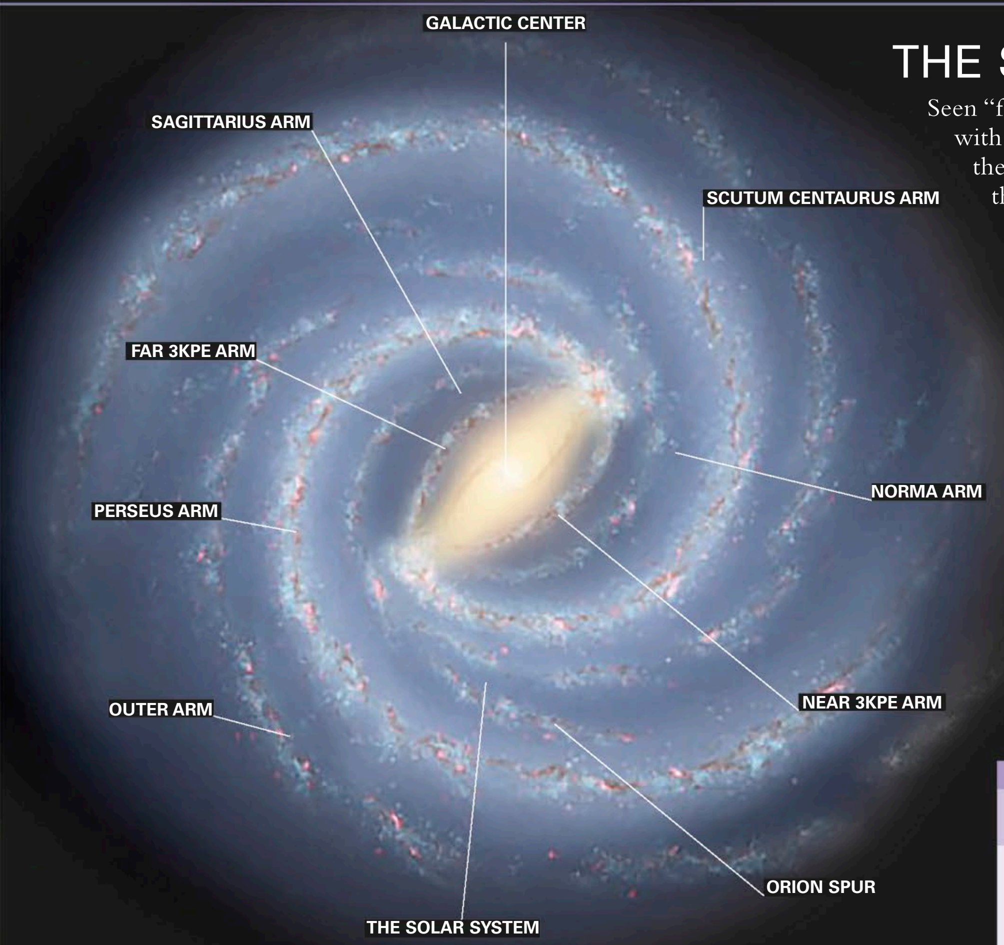
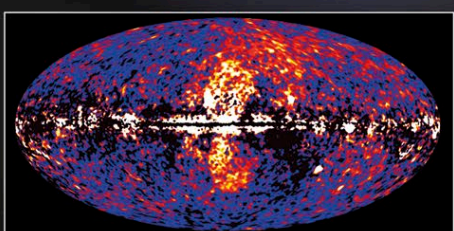


THE SPIRAL ARMS

Seen "face-on," the Milky Way would look like a huge catherine wheel, with the majority of its light coming from the arms spiraling out from the ends of a bar of stars that extends from the central bulge. In fact, the material in the spiral arms is generally only slightly denser than the matter in the rest of the disk. It is only because the stars that lie within them are younger, and therefore brighter, that the pattern in spiral galaxies shows up. Two mechanisms are thought to create the Milky Way's spiral structure. Density waves, probably caused by gravitational attraction from other galaxies, ripple out through the disk, creating waves of slightly denser material and triggering star formation (see pp.238–239). By the time the stars have become bright enough to etch out the spiral pattern, the density waves have moved on through the disk, starting more episodes of star formation and leaving the young stars to age and fade. High-mass stars eventually explode as supernovae, sending out blast waves that also pass through the star-making material, triggering further star formation.

SPINNING GALAXY
The galaxy rotates differentially—the closer objects are to the center, the less time they take to complete an orbit. The Sun travels around the galactic center at about 500,000 mph (800,000 kph), taking around 225 million years to make one orbit.



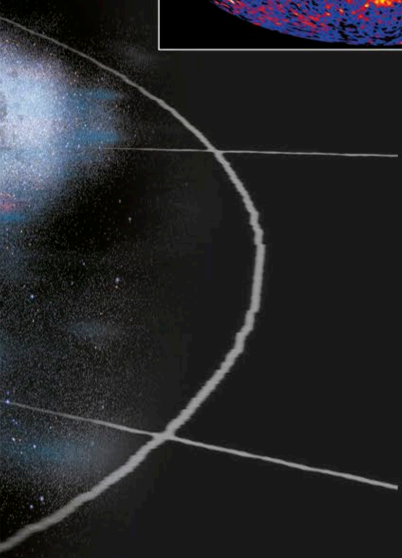
GAMMA-RAY BUBBLES
Two huge bubbles of gamma-ray-emitting gas fill the space above and below the Milky Way's hub. They may be remnants of past activity in the central black hole.

MYTHS AND STORIES

HEAVENLY MILK

There are many myths involving the formation of the Milky Way. In Greek mythology, Hercules was the illegitimate son of Zeus and a mortal woman, Alcmene. It was said that when Zeus's wife, while suckling Hercules, heard he was the son of Alcmene, she pulled her breast away and her milk flowed among the stars.

MYTH IN ART
The Origin of the Milky Way (c.1575) by Jacopo Tintoretto was inspired by the Greek myth.



MAPPING THE MILKY WAY
The Milky Way's structure is defined by its major arms, each named after the constellation in which it is most prominent—the brightest arm is that in Sagittarius, beyond which lies the galactic nucleus. The solar system lies near the inner edge of the Orion Arm. All the arms lie in a plane defined by the galactic disk. The nucleus forms a bulge at its center, and globular clusters orbit above and below it in the halo region.

STELLAR POPULATIONS

Stars are broadly classified into two groups, called populations, based on age and chemical content. Population I consists of the youngest stars, which tend to be richer in heavy elements. These elements are primarily produced by stars, and Population I stars are created from materials shed by existing stars. In the Milky Way, the majority of Population I stars lie in the galactic disk, where there is an abundance of star-making material. Population II stars are older, metal-poor stars, existing primarily in the halo, but also in the bulge. Most are found within globular clusters, where all star-making materials have been used up and no new star formation is taking place.

STAR MOTION
The stars in the bulge have the highest orbital rates. They can travel thousands of light-years above and below the plane of the Milky Way. Within the disk, stars stay mainly in the plane of the galaxy as they orbit the galactic center. Stars in the halo plunge through the disk, reaching distances many thousands of light-years above and below it.